Computer Networks Notes

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1. Network Fundamentals

- A network is an interconnected system of independent computing devices, known as nodes, linked via wireless or wired channels. Its primary purpose is to enable communication and the sharing of data, services, and resources.
- **Core Elements:**
  - **Nodes:** Hardware devices such as PCs, routers, and servers.
  - **Links:** The physical or wireless medium, including Wi-Fi, Ethernet, and fiber optics.
  - **Protocols:** The established rules that dictate data exchange, such as HTTP and TCP/IP.
- **Classification:** Networks are generally categorized by their operational scale. Common types include LAN (Local Area Network), MAN (Metropolitan Area Network), and WAN (Wide Area Network).

2. Understanding IP Addresses

- An Internet Protocol (IP) address is a logical identifier functioning at the Network Layer (Layer 3). It provides both host identification and location addressing capabilities.
- **Main Versions:**
  - **IPv4:** Utilizes a 32-bit dotted-decimal format (e.g., 192.168.1.1) and offers approximately 4.3 billion unique addresses.
  - **IPv6:** Developed to resolve the depletion of IPv4 addresses, it uses a 128-bit hexadecimal format.
- **Behavior:** IP addresses can be static or dynamic. Unlike hardware addresses, an IP address changes whenever a device connects to a new network.

3. Understanding MAC Addresses

- A Media Access Control (MAC) address is a permanent, physical identifier burned into a device's Network Interface Card (NIC) by its manufacturer. It operates at the Data Link Layer (Layer 2).

- **Structure:** It is a 48-bit address formatted as 12 hexadecimal digits.
  - The first 24 bits represent the Organizationally Unique Identifier (OUI), which identifies the hardware manufacturer.
  - The final 24 bits serve as the specific serial number for the device.
- **Function:** While IP addresses route packets between different networks, MAC addresses ensure frames are delivered to specific devices on a local network segment.

4. The Origins of the OSI Model

- In the 1970s and 1980s, network architectures were strictly proprietary. Equipment from different manufacturers, such as IBM and DEC, could not communicate natively, making integration either impossible or extremely costly.
- In 1984, the ISO introduced the Open Systems Interconnect (OSI) model to solve this vendor lock-in.
- **Purpose:** It established open standards for interoperability, standardized interfaces, and modularized network development by splitting complex communication processes into seven manageable layers.

5. The 7 OSI Layers Breakdown

(Mnemonic: All People Seem To Need Data Processing)

- **Layer 7: Application Layer**
  - **Data Unit:** Data.
  - **Function:** Directly provides network services to user applications, handling resource identification and service advertisement.
  - **Protocols:** DNS, SMTP, FTP, HTTP, HTTPS.
- **Layer 6: Presentation Layer**
  - **Data Unit:** Data.
  - **Function:** Acts as a network translator. It manages data compression, translation (e.g., ASCII to EBCDIC), and encryption/decryption (SSL/TLS).
- **Layer 5: Session Layer**
  - **Data Unit:** Data.
  - **Function:** Controls communication dialogues by establishing, managing, and terminating sessions. It oversees authentication, authorization, and checkpointing for interrupted transfers.
- **Layer 4: Transport Layer**
  - **Data Unit:** Segments (TCP) / Datagrams (UDP).

    - **Function:** Guarantees error-free, end-to-end data delivery. It handles segmentation, reassembly, flow control, and error control.
- **Layer 3: Network Layer**
    - **Data Unit:** Packets.
    - **Function:** Manages logical addressing and determines the optimal path for moving data across multiple networks via routers.
- **Layer 2: Data Link Layer**
    - **Data Unit:** Frames.
    - **Function:** Handles node-to-node delivery within a local network using physical MAC addressing. It also manages framing and media access control.
- **Layer 1: Physical Layer**
    - **Data Unit:** Bits.
    - **Function:** Transmits raw, unstructured bits over physical mediums like cables and radio waves, defining data rates and electrical voltages.

6. PDUs: Segments, Packets, and Frames

- Protocol Data Units (PDUs) are layers of data wrapped in specific header information as they move through the OSI model.
- **Segment (Layer 4):** The Transport Layer chunks application data into segments. The header includes sequence numbers and source/destination port numbers.
- **Packet (Layer 3):** The Network Layer encapsulates the segment into a packet by adding a routing envelope that contains the source and destination IP addresses.
- **Frame (Layer 2):** The Data Link Layer wraps the packet in a frame. It adds source/destination MAC addresses to the header and a Cyclic Redundancy Check (CRC) to the trailer to detect physical transmission errors.

7. Transport Protocols: TCP, UDP, and QUIC

- **TCP (Transmission Control Protocol):** A connection-oriented, reliable protocol that guarantees data is delivered in order. It uses "Windowing," a flow-control mechanism where the receiver dictates how much data the sender can transmit before requiring an acknowledgment, preventing buffer overflow.
- **UDP (User Datagram Protocol):** A connectionless, unreliable, "fire and forget" protocol. It has low overhead and is used when speed is prioritized over perfect delivery, such as in VoIP or live streaming.

- **QUIC:** A modern transport protocol developed by Google that runs on top of UDP. It combines the speed of UDP with the reliability of TCP, offering a 0-RTT handshake, preventing head-of-line blocking, and allowing seamless connection migration.

8. The TCP Three-Way Handshake

- TCP uses a three-step handshake to synchronize sequence numbers and establish a reliable connection.
  1. **SYN:** The client sends an Initial Sequence Number (ISN) to request a connection.
  2. **SYN-ACK:** The server acknowledges the request and responds with its own ISN.
  3. **ACK:** The client acknowledges the server's sequence number, finalizing the connection.
- Acknowledgment numbers always represent the next expected sequence number (e.g., X+1).

9. Sockets and Well-Known Ports

- A **Socket** is defined as an IP Address + Port Number + Protocol (TCP/UDP).
- **Essential Ports:**
  - **20/21:** FTP (File Transfer).
  - **22:** SSH (Secure Shell).
  - **23:** Telnet (Terminal Emulation).
  - **25:** SMTP (Email Routing).
  - **53:** DNS (Domain Name Resolution).
  - **80:** HTTP (Web Traffic).
  - **443:** HTTPS (Secure Web Traffic).
  - **110:** POP3 (Email Retrieval).
  - **143:** IMAP (Email Sync).
  - **161:** SNMP (Network Management).
  - **3306:** MySQL.
  - **5432:** PostgreSQL.

10. Hardware Ecosystem

- **Hub (Layer 1):** An unintelligent legacy device operating in half-duplex that blindly broadcasts incoming signals to all ports.
- **Repeater (Layer 1):** Regenerates and amplifies signals to prevent attenuation over long distances.

- **Switch (Layer 2):** An intelligent device that maintains a MAC Address Table to direct traffic to specific ports within a LAN, allowing for full-duplex communication.
- **Bridge (Layer 2):** Connects two distinct local network segments by filtering MAC addresses.
- **Router (Layer 3):** Connects separate networks together by reading IP addresses and utilizing routing tables to determine the best path.
- **Gateway:** Acts as a protocol translator to connect completely different network architectures.
- **Firewall:** Filters incoming and outgoing traffic based on security rules using stateless, stateful, or deep packet inspection methodologies.
- **Access Point (Layer 2):** Bridges wireless radio waves to a wired Ethernet network.

11. DHCP Workflow

- **Dynamic Host Configuration Protocol (DHCP)** is a Layer 7 protocol (running on UDP ports 67/68) that automates the assignment of network configurations to prevent IP conflicts.
- **The DORA Process:**
  1. **Discover:** The client broadcasts a request to find a DHCP server.
  2. **Offer:** A server reserves an IP and sends a unicast offer to the client.
  3. **Request:** The client broadcasts its acceptance of the offered IP.
  4. **Acknowledge:** The server sends a final unicast confirmation.
- **Provided Parameters:** DHCP grants devices an IP Address, Subnet Mask, Default Gateway, and DNS Server IP.
- **APIPA:** If a device cannot reach a DHCP server, it assigns itself an Automatic Private IP Address (APIPA) in the 169.254.X.X range, allowing only local network communication without internet access.